

# Gilberdi Axel Nathaniel Sinaga

<https://gilberdi.vercel.app>

Email: [gilberdising012@gmail.com](mailto:gilberdising012@gmail.com)

Mobile: +62-812-6248-1507

## SUMMARY

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Final-year computer science student with a passion for software engineering and a focus on building efficient and scalable systems. Demonstrates a strong ability to contribute in professional environments and adapt quickly to new technical challenges. Committed to writing clean, maintainable code and continuously expanding expertise across the broader software landscape.

## EDUCATION

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- **Universitas Sumatera Utara** Medan, North Sumatra, Indonesia  
*Bachelor of Science in Computer Science; GPA: 3.97*  
June 2021 – Expected November 2026
- **Universiti Teknologi MARA** Shah Alam, Selangor, Malaysia  
*Student Exchange Program; GPA: 3.78*  
March 2023 – August 2023

## EXPERIENCE

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- **Systatum** Yokohama, Kanagawa, Japan (Remote)  
*Intern Member of Technical Staff*  
May 2025 - November 2025
  - **Gara**: Researched and implemented an interpreter in **Go** using **ANTLR4** for Gara, a programming language designed to help Indonesian students learn to code through familiar, native-language syntax.
  - **Workaty**: Led DevOps infrastructure setup using **Capistrano** and **Microsoft Azure** for Workaty, an HR operating system. Developed **rbacr**, and integrated it into the backend. Contributed to backend feature development in **Crystal** with **PostgreSQL**, and collaborated with the frontend team to align API contracts and ensure smooth integration across the stack.
- **PT Smut Tech Indonesia** Medan, North Sumatra, Indonesia  
*Software Engineer (Internship)*  
July 2024 - October 2024
  - **ERP-VAS**: Created a customer and prospect map view with Leaflet library and GeoJSON data in **Laravel** on an Enterprise Resource Planning (ERP) application for a company. Built charts for statistics using Metronic.
  - **SMUT SaaS**: Developed frontend for a SaaS application using **Next.js** and **Tailwind CSS**.
  - **KOMINA**: Developed a mobile application using **Flutter** and **gRPC** for a fisheries cooperative. Involved in the development of some features of the ERP web application in **Laravel**.
- **PT Pharos Indonesia** Jakarta Selatan, Jakarta, Indonesia  
*Back End Developer (Internship)*  
February 2024 - June 2024
  - **Century Marketplace**: Involved in the development of REST API in **Go** using **Cloud Firestore** as the database for a pharmaceutical e-commerce application serving 400+ partner pharmacies. Implemented a notification feature for the mobile application using **Firebase Cloud Messaging (FCM)**.
- **Azura Labs** Semarang, Central Java, Indonesia (Remote)  
*Back End Developer (Internship)*  
August 2022 - December 2022
  - **WoiShop**: Built a mock-up e-commerce REST API using **Express.js** and **PostgreSQL** in a training.

## PROJECTS

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- **rbacr**: The first Role-Based Access Control Library for the Crystal Programming Language. Used in production in Workaty's backend.  
Technologies: Crystal
- **Decodream**: IC-blockchain-based platform that analyzes and transforms dreams into digital art using AI technology. Submitted to Codefest Hackathon 11.  
Technologies: React.js, Motoko, Sass
- **Last Boxd**: Generates highly customizable film collages from Letterboxd data, featuring 30+ aesthetic controls, AI assistants, and cross-device cloud sync. Refactored the rendering pipeline from a dual interactive/export mode into a fully WYSIWYG canvas editor, and optimized performance for smooth real-time previews at high-resolution. As of July 2026, the site attracts 200+ monthly visitors and secured its first paid PRO subscriber in June 2026.  
Demo: <https://youtu.be/JUEiooj9VGQ>  
Technologies: Go, Svelte, Firebase

## SKILLS

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- **Frameworks:** Express.js, Laravel, Flutter, React.js, Next.js, Svelte, Tailwind CSS, Django
- **Programming Languages:** JavaScript, TypeScript, Ruby, Crystal, Go, Dart, Python, PHP, Bash
- **Databases:** PostgreSQL, MySQL, SQLite
- **Tools:** macOS, Visual Studio Code, Neovim, Git, Node.js, Microsoft Azure